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The impact of SME's pre-bankruptcy financial distress on earnings management tools

Domenico Campa [a,*], María-del-Mar Camacho-Miñano [b]

[a] School of Business, Trinity College Dublin, College Street, Dublin, Ireland
[b] Department of Financial Economics and Accounting II (Accounting), School of Business Administration and Economics, Complutense University of Madrid, 28223 Madrid, Spain

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ABSTRACT

Previous literature finds that situations that put managers under significant levels of pressure (e.g. IPO, upcoming credit rating changes, violation of debt covenant, etc.) might affect the way earnings are manipulated. The aim of this study is to investigate whether the pressure caused by the non-temporary level of financial distress, conditions the choice between real activity and discretionary accrual manipulation. Using a selection of financial distress indexes and pre-bankruptcy data of a sample of bankrupt small companies operating in a code law country, our findings show that, on average, firms with higher levels of financial distress show more extensive signs of upward earnings management through real transaction manipulation rather than accruals and *vice versa*. Accordingly, real activity earnings management is preferred over accruals when managers are under significant levels of 'pressure' such as being close to face a bankruptcy procedure, despite its implications for the firm in the long term.

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1. Introduction

Managers have different incentives to manipulate earnings that may affect the quality of financial statements and alter their reliability. Especially during a financial crisis, the pressure brought by poor results and/or tough financial situations is an 'ideal' context for managers to expand the alteration of the current performance of firms. For example, extant literature indicates that managers have stronger motivations for manipulating reported earnings during financial distress situations such as the period preceding bankruptcy procedures (e.g. Adu-Boateng, 2011; Burgstahler & Dichev, 1997). Indeed, bankrupt companies manipulate earnings upwards in comparison with healthy firms (Campa & Camacho-Miñano, 2014; Rosner, 2003) and, in some cases, entities do that to 'clean' the negative signals of financial distress (Iatridis & Kadorinis, 2009; Jaggi & Lee, 2002; Sweeney, 1994).

Recent studies (Choi, Kim, & Lee, 2011; Trombetta & Imperatore, 2014) have also analyzed the effect of economic downturns on financial reporting quality since the former enlarge the number of firms in financial distress situations, especially among small companies. They show that earnings management increases when an economic crisis becomes worse. Hence, it is interesting to analyse the impact of economic downturns at a microeconomic level, i.e. the index of financial distress of each firm, on earnings quality through the study of the two tools of earnings management, real activity and accrual manipulation (Schipper, 1989). Literature that investigates what drives the choice between these two earnings management tools indicates that real activity manipulation is chosen in the presence of 'extreme' circumstances, when managers need immediate results, regardless of the disadvantages and the costs it has for companies in the long term (e.g. Chamberlain, Butt, & Sarkar, 2014; Kim, Kim, & Song, 2013). Accordingly, we aim to investigate whether the earnings management tools used by managers of companies that are experiencing 'extreme' financial situations also depend on the level of firms' financial distress. Indeed, there is evidence that distressed firms behave differently from others, for example, when looking at their financial structure decisions (Pindado, Rodrigues, & de la Torre, 2006).

Our investigation focuses on a sample of small and medium companies (SME) which have been involved in a bankruptcy procedure and operate in a code law country, Spain. The analysis of small companies would allow us to focus on the most relevant part of the economy of developed countries. Indeed, according to latest available statistics, 99.8% of firms in Europe, in 2013, were SMEs; they employed 66.8% of workers and delivered 57.9% of gross value added generated by the

* Corresponding author at: Trinity College Dublin, School of Business, College Street, Dublin 2, Ireland. Tel.: +353 1 8963775; fax: +353 1 6799503.
E-mail address: domenico.campa@hotmail.com (D. Campa).

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private, non-financial, business sector (European Commission, 2014, p. 10). In addition, SMEs are those that suffer the most during financial distress given their limited choices in terms of access to funds (Dewaelheyns & Van Hulle, 2008). On the other hand, the use of a code law setting would allow us to investigate a context usually less explored by the extant literature. Indeed, the majority of studies is 'biased' towards common law countries and listed companies while evidence indicates that earnings manipulation is different and more pervasive in code law institutional settings (Leuz, Nanda, & Wysocki, 2003) and among non-listed entities (Ball & Shivakumar, 2005). The incentive for the use of different tools of earnings manipulation could change depending on the institutional setting. The more extensive use of accounting data in contracting in code law countries may motivate managers to manipulate accruals more than in common law settings (e.g. Arnedo, Lizarraga, & Sánchez, 2007; Nabar & Boonlert-U-Thai, 2007). In common law countries investors ask for higher quality financial statements and companies should live up to the market's expectations thus managers have more incentives to use real activity manipulation instead of accrual management because the former is less detectable by external stakeholders (Enomoto, Kimura, & Yamaguchi, in press). If we want to generalize research results, it would be important to test different contexts and samples. Moreover, Spain is a country where the outcome of a bankruptcy procedure is quite hard on companies. Evidence indicates that around 90% of firms filing for bankruptcy procedure, in Spain, are liquidated[1], with negative consequences for managers and all other firms' stakeholders. Given this statistic, managers are aware that an involvement in a bankruptcy procedure basically means the end of their company and a failure in their job. For all these reasons, in line with other studies concentrated on common law contexts and listed companies, we investigate whether Spanish SMEs in a severe non-temporary financial distress situation have more incentives for using real activity manipulation than discretionary accruals.

The main contribution of the paper is that evidence highlights that the level of pre-bankruptcy financial distress does affect managers' behaviour. We find that, on average, before bankruptcy, firms with higher non-temporary levels of financial distress manage earnings upwards using real transaction manipulation more than other lower-distressed bankrupt companies. These firms also exhibit less income-increasing accrual manipulation than bankrupt companies with lower pre-bankruptcy levels of financial distress. It would support the view that real activity manipulation is seen as the most 'reliable' and 'effective' tool to be used when the events require some 'drastic' and 'ultimate' actions. Moreover, although other studies found that real activity manipulation is less extensive than accrual manipulation in code law contexts (Enomoto et al., in press; Leuz et al., 2003), our evidence indicates that the pressure caused by a situation of financial distress might be even more relevant than the institutional setting when investigating managers' behaviours and the quality of information.

The rest of the paper is organized as follows. Section 2 frames this study within the literature that deals with the trade-off between real activity and accrual manipulation and presents the main hypothesis. Section 3 details the main distress indexes and the methodology used to gather evidence in order to test our hypothesis. It also describes the sample selection procedure. Section 4 presents and discusses the empirical results and, finally, Section 5 concludes this research highlighting its main implications and limitations.

2. Background and hypothesis development

Real activities and accruals are the two main tools available to managers in order to manipulate earnings. Gunny (2010) states that managers may prefer real activity management over accruals manipulation because, for example, accounting standards setters are making accruals more difficult to manipulate and because auditors exercise a closer monitoring over accruals. In addition, Graham, Harvey, and Rajgopal (2005) find that managers would rather take economic actions with future persistent consequences than focusing on accounting choices whose effects are transitory. On the other hand, Zang (2012) suggests that managers may prefer accrual management to real activity manipulation because the former can take place after the fiscal year-end, when they actually know if earnings management is needed. Yang, Rahman, and Bradbury (2015) ascertain that the use of the two tools of earnings management depends on the cost and the circumstances that incentivise firms to manipulate earnings. As a consequence, the implications of the choice between these earnings management tools are crucial for firms' evolution. Real activity manipulation has greater efficacy than accruals because it is easier to implement and it is also more difficult to detect by others due to its opacity (García Lara, García Osma, & Mora, 2005; Kothari, Roychowdhury, & Mizik, in press) even if it produces negative effects on future cash flow (Chen, Yen, & Chang, 2009), profitability and stock price (Bhojraj, Hribar, Picconi, & McInnis, 2009).

Historically, the discussion on earnings management primarily focused on accrual manipulation strategies and there is a large volume of studies devoted to documenting accrual earnings management (e.g. Nelson, Elliot, & Tarpley, 2002; Teoh, Welch, & Wong, 1998) while real actions undertaken to manage earnings have received less attention by the accounting literature (Xu, Taylor, & Dugan, 2007; Zang, 2012).

The transition from accrual to real activity manipulation could be conditioned by the role of regulators and auditors, who cannot restrain firms from engaging in real activity manipulation but could strength the accounting regulation, thus limiting the use of accruals (Enomoto et al., in press).

Real activity studies are based on firms' choices that affect the real conduct of the business and its cash-flows, for example reducing research and development expenses (i.e. Cheng, 2004; Dechow & Sloan, 1991), altering production output (e.g. Baber, Fairfield, & Haggard, 1991; Perry & Grinaker, 1994), cutting advertising expenditures (Cohen, Mashruwala, & Zach, 2010), increasing revenues by offering large discounts or unusually favourable credit terms to customers (e.g. Jackson & Wilcox, 2000; Roychowdhury, 2006), selling non-current assets (e.g. Bartov, 1993; Herrmann, Inoue, & Thomas, 2003).

Currently, there is a new research line, mainly still in the form of working papers, that is looking at the driving forces that affect managers' choice between real activity and accrual earnings management and this trade-off is also the focus of this paper. We classified, in Table 1, all these studies into two big groups: firms' external and internal factors that condition the preference for one earnings management strategy rather than the other. Firstly, we show the situations that condition the choice between accruals and real activity in each paper, then we highlight how their findings are not conclusive and, finally, we discuss the motivations for real activity manipulation instead of accrual management and the consequences for companies.

In the first group, we find papers that relate the choice of one particular earnings management tool on the basis of external firms' factors such as the introduction of new regulations like the Sarbanes and Oxley Act (Cohen, Dey, & Lys, 2008), IFRS (Ipino & Parbonetti, 2011), strengthened accounting standards (Ewert & Wagenhofer, 2005), the fiscal and legal regime of the countries in which firms operate (Durnev, 2010; Enomoto et al., in press), firm's stock prices (Badertscher, 2011), audit quality (Burnett, Cripe, Martin, & McAllister, 2012), external monitoring (Wongsunwai, 2012), the effect of regulated market and market failure risk on initial public offering (Alhadab, Clacher, & Keasey, 2013a, 2013b) and upcoming credit rating changes (Kim et al., 2013).

In the second group, we have evidence that explains the shift from real activity to accrual manipulation or *vice versa* in the presence of firms' internal situations such as income smoothing (Matsuura, 2008), meeting earnings benchmarks (Bjurman & Weihagen, 2013; Zang,

[1] Source: INE – Spanish National Statistics Institute, 2011.

Table 1
Classification of papers related to real activity vs accrual earnings management.

Classification criteria		Authors	Sample (Country/Type of firms)	Earnings management 'strategy'	Research key results
EXTERNAL FACTORS OF THE FIRM	After accounting standards are strengthened	Ewert and Wagenhofer (2005)	USA/Listed	AM>REM	Tighter accounting standards increase AM
	After the Sarbanex and Oxley Act (SOX)	Cohen et al. (2008)	USA/Listed	REM>AM	Managers increased REM and decreased AM after the adoption of SOX although benchmarks are important incentives to do it more or less.
	Lower taxation, looser legal regime and less stable country	Durnev (2010)	Different countries/Listed	AM>REM	Firms that have headquarters registered in OFCs with a higher Offshore Attitude Index prefer AM to increase their reported earnings but use REM as a supplement
	Overvaluation of firm's stock prices	Badertscher (2011)	USA/Listed	REM>AM	Managers use AM in early years but change to REM in order to sustain their overvalued equity.
	After the mandatory IFRS adoption	Ipino and Parbonetti (2011)	37 countries/Listed	REM>AM	Firms decrease in AM and increase in REM for mandatory IRFS adoption only in countries with strict enforcement regimes
	Audit quality	Burnett et al. (2012)	USA/Listed	REM>AM	Firms with high audit quality are more likely to use REM and less AM to meet their target
	External monitoring	Wongsunwai (2012)	USA/Listed	REM=AM	After going public, companies backed by higher quality VCs have lower abnormal accruals, lower earnings management through REM
	1) Great scrutiny from auditors and regulators 2) Market-leader status 3) High marginal rates	Zang (2012)	USA/Listed	REM=AM	REM and AM are substitute. Firms under higher auditor and regulators scrutiny reduce AM. Firms that are market leader prefer REM to AM
	IPO and lightly regulated market	Alhadab et al. (2013a)	UK/Listed	REM>AM	IPO firm manage earnings upwards by manipulating REM more extensively than AM in lightly regulated market
	IPO failure risk	Alhadab et al. (2013b)	UK/Listed	REM=AM	Firms engaged in higher level of REM and AM exhibit high probability of IPO failure and low survival rates
	Upcoming credit rating changes	Kim et al. (2013)	USA/Listed	REM>AM	Firms with upcoming credit rating changes are more engagement in REM than in AM
	Strong investor protection	Enomoto et al. (in press)	38 countries	REM>AM	REM is implemented in the countries with stronger investor protection
INTERNAL FACTORS OF THE FIRM	Income smoothing	Matsuura (2008)	Japan/Listed	REM=AM	Relation between real and accounting earnings management is sequential and managers use complementarily them to smooth income
	Around seasoned equity offerings	Cohen and Zarowin (2010)	USA/Listed	REM=AM	SEO firms use real, as well as accrual-based, earnings management methods as a trade-off depending on their ability to use accruals and its costs
	CEO's age	Demers and Wang (2010)	USA/Listed	AM>REM	Young executives use AM instead of REM and even manipulate less than their counterparts
	To target zero	Eisele (2012)	Germany/Listed	AM>REM	German executives of firms with small profit use AM instead of REM
	SEO valuation	Kothari et al. (in press)	USA/Listed	REM>AM	Managers engage in RAM at the time of SEO due to the lower risk of detection even though RAM is more costly in the long run
	1) Low accounting flexibility 2) High institutional ownership 3) Poor financial health	Zang (2012)	USA/Listed	1)AM>REM 2)REM>AM	1)and 2) firms with high institutional investors and higher accounting flexibility prefer AM to REM 3)Healthier firms prefer REM to AM
	1) Surrounding the debt covenant violation 2) Quarter of the debt covenant violation	Chamberlain et al. (2014)	USA/Listed	1) AM = REM 2) REM >AM	Both manipulation tools in order to avoid violation of debt covenant but AM surrounding and REM in the quarter of the violation
	Offering SEO	Bhuiyan and Henry (2013)	Australia/Listed	REM=AM	Managers of Australian SEO firms engage both REM and AM in the SEO-years.
	Specific earnings benchmarks	Bjurman and Weihagen (2013)	22 countries/Listed	REM>AM	Managers preferring REM over AM to reach earnings benchmarks
	CEO turnover	Geertsema et al. (2014)	USA/ Listed	REM>AM	New CEOs manipulate REM instead of AM to manage earnings downwards
	CEOs with more financial experience	Jiang et al. (2013)	China/Listed	AM>REM	CEO's with more financial experience tend to do less REM but no evidence of more or less AM
	Internal control weaknesses (ICW)	Lenard et al. (2013)	USA/Listed	REM>AM	ICW-firms are more prone to using REM than AM
	Ownership structure	Achleitner et al. (2014)	Germany/non-Listed	AM>REM	Family firms are more engagement in AM and less in REM than their counterparts.

In **bold** are the papers that are already published.*Zang (2012)'s paper has several hypotheses so we divided them according to our classification.
AM = Accrual manipulation; REM = Real activity earnings manipulation.

2012), or zero earnings (Eisele, 2012), seasoned equity offerings (SEO) or going for IPO (Bhuiyan & Henry, 2013; Cohen & Zarowin, 2010; Kothari et al., in press), avoiding debt covenant violations (Chamberlain et al., 2014) or having internal control weaknesses (Lenard, Petruska, Alam, & Yu, 2013). Other studies also relate the choice towards one earnings management tool over the other in relation to the ownership structure of firms (Achleitner, Guenther, Kaserer, & Siciliano, 2014), financial experience, turnover and age of CEOs (Demers & Wang, 2010; Geertsema, Lont, & Lu, 2014; Jiang, Zhu, & Huang, 2013).

It is interesting to point out that the results about earnings manipulation strategies reported by these studies are not conclusive. There are papers that provide evidence of a shift from accruals to real activity manipulation (e.g. Burnett et al., 2012; Cohen & Zarowin, 2010; Cohen et al., 2008; Kim et al., 2013) but there are also studies that highlight the opposite strategy (e.g. Achleitner et al., 2014; Eisele, 2012) and even others that show that the two earnings management tools are substitutes depending on managers' targets, costs and timing (e.g. Alhadab et al., 2013b; Zang, 2012). The reason is that, as discussed before, each earnings manipulation tool has different effects and consequences.

From the analysis of the previously mentioned studies, we find that some of them indicate that the presence of 'stressful situations', where managers need immediate results to achieve specific benchmarks or goals, is one of the factors that give rise to a preference over one of the two earnings management tools (Bjurman & Weihagen, 2013). More precisely, in these scenarios, firms opt for more real activity manipulation and less accrual management in comparison with other companies. Another factor of pressure for managers is the non-temporary or permanent financial distress. While there are already studies focused on earnings management among financial distressed companies (e.g. Campa & Camacho-Miñano, 2014; Charitou, Lambertides, & Trigeorgis, 2007a, 2007b; García Lara, García Osma, & Neophytou, 2009; Habib, Uddin Bhuiyan, & Islam, 2013; Rosner, 2003), they mainly show that bankrupt firms use accruals and real activities to manipulate earnings upwards the years prior entering into a legal bankruptcy procedure in comparison with healthy companies and regardless of the institutional setting in which they operates. However, to best of our knowledge, they do not analyse whether the severity of the pre-bankruptcy financial distress might have an influence on the tools used by the management to carry out such an upward earnings management. In regard of this, we believe that managers of bankrupt firms with higher pre-bankruptcy levels of financial distress feel more pressure to use more drastic actions in order to try to avoid all the negative consequences bankruptcy can cause. More specifically, because of the higher probability of being involved in a legal procedure for bankruptcy, they are more likely to manage earnings through real activity manipulation. Indeed, this strategy is easier to implement and has less risk of litigation (Graham et al., 2005) because is more difficult to detect by the auditors (Eisele, 2012; Kothari et al., in press; Zang, 2012) than accrual manipulation thus it could preserve the confidence of creditors and/or account advisors who may identify window dressing activities. In addition, real activity management can also provide distressed firms with quick cash because it could come from large discounts given to customers (Roychowdhury, 2006) or from the sale of companies' assets (Blazy, 1999). At this stage is also worth noticing that, while Zang (2012) indicates that healthier firms prefer real activity earnings management rather than accrual manipulation, she does not use companies that filed for bankruptcy so the pressure brought by the financial distress in her scenario is different from that of companies that, in the end, filed for bankruptcy.

Bearing all this in mind, our hypothesis can be stated as follows:

H. *Higher pre-bankruptcy levels of financial distress result in more pervasive income-increasing earnings management through real transaction (either sale or cost) manipulation than through accrual manipulation.*

3. Methodology

3.1. Measuring financial distress

The paper aims to explore the relation between earnings management behaviours and pre-bankruptcy level of financial distress of bankrupt firms, therefore the estimation of the latter variable is of particular relevance. The concept of distress has been arbitrary and heterogeneous in prior literature (Balcaen & Ooghe, 2006; Laitinen, 2011). In addition, since the first index for financial distress implemented by Beaver (1966), many other proxies have been developed (see Altman, 1968, 1993; Carcello & Neal, 2000; Fich & Slezak, 2008; Jaggi & Lee, 2002; Mousavi, Ouenniche, & Xu, 2015; Tinoco & Wilson, 2013; Zmijewski, 1984). Having distress indexes developed and used to predict bankruptcy, they provide us with a tool that helps us in ranking the companies on the basis of the severity of their level of distress and, consequently, on the probability that they could be involved in a legal procedure for bankruptcy.

Among all the above mentioned indexes, our main analyses employ the methodology developed by Fich and Slezak (2008), being it based exclusively on accounting data, thus applicable to our sample that includes non-listed enterprises[2]. Fich and Slezak (2008) indicator of distress is a dummy variable that takes the value of 1 if the ratio between operating profit and interest expenses is below 1 (i.e. evidence of financial distress) and zero otherwise. It is based on the assumption that if a company is not able to generate an operating profit large enough to cover its interest expenses, it could be soon facing default on its debt. Since Fich and Slezak (2008) proxy of distress is based on an accounting ratio rather than a score, its use also addresses the concern that distress indicators derived from scores might be ambiguous, lead to spurious results and are usually more effective in predicting other companies' problems rather than the actual bankruptcy (Grice & Ingram, 2001).

3.2. Estimation of earnings management

3.2.1. Real earnings management proxies

For the selection of our real earnings manipulation proxies, we follow Roychowdhury's (2006) methodology[3] and focus on sale and production cost manipulation as they usually are the most significant accounts in a firm's annual report and revenues, receivables, cost of goods sold and inventory are the most frequently managed items (Dechow, Ge, Larson, & Sloan, 2011; Ricci, 2011) [4].

First of all, we analyse sale manipulation estimating the following cross-sectional regression (1), for each combination of industry[5] and

[2] Although the Altman score is the most commonly used index by studies focused on bankruptcy, García Lara et al. (2009) in their footnote 9 state that "Altman's (1968) Z-score model is built using LDA, a technique that makes strong demands on the structure of data. First, it requires that the financial ratios are normally distributed. (…) Also, the ratios of failed companies should have the same variance-covariance structures as those of continuing firms (…)".

[3] Roychowdhury (2006) methodology is relevant for this paper as it has been developed for the analysis of distressed companies. For this reason we can be reasonably sure that it is robust for the analysis of firms in 'extreme' operating situations such as those included in our sample.

[4] They are the most suitable for the kind of companies we are investigating as those proxies have already been employed in the analysis of firms in financial distress and in bankruptcy (e.g. García Lara et al., 2009; Zang, 2012), during economic crisis (e.g. Trombetta & Imperatore, 2014) and in countries different from that where the methodology was developed (e.g. Chen et al., 2009; García Lara et al., 2009).

[5] We group the companies into four industries – 'Construction', 'Industrial', 'Services' and 'Wholesalers' – on the basis of the INE (National Statistics Institute) classification.

year identified, to calculate the normal level of cash flow given reported sales (Roychowdhury, 2006)[6].

$$CFO_{it}/ASSETS_{t-1} = \alpha + \beta_1(1/ASSETS_{it-1}) + \beta_2(SALES_{it}/ASSETS_{it-1}) + \\ + \beta_3(\Delta SALES_{it}/ASSETS_{it-1}) + \varepsilon_{it} \quad (1)$$

Where:

CFO	Cash flow from operations (calculated as earnings before interests and taxes + depreciation and amortization, +/- changes in inventories, changes in trade and other receivables, changes in trade and other payables).
ASSETS	Total assets.
SALES	Net sales.
ΔSALES	Change in net sales.

For each firm, abnormal cash flow (*ABNCFO*) is obtained as the difference between the actual *CFO* and the 'normal' *CFO* calculated using the estimated coefficients from the above Eq. (1).

The manipulation of production costs is analysed by the following model (2) which is used to estimate the normal level of production costs (Roychowdhury, 2006). The latter are defined as 'the sum of the cost of goods sold and change in inventory as it overcomes the disadvantages that would have been observed by using the value of the cost of goods sold alone', as explained in detail by Roychowdhury (2006, p. 339). The same definition of production cost is also used by Gunny (2010) who explicitly highlights the several benefits of this approach[7]. The model is estimated separately for each combination of industry and year.

$$PROD_{it}/ASSETS_{it-1} = \alpha + \beta_1(1/ASSETS_{it-1}) + \beta_2(SALES_{it}/ASSETS_{it-1}) + \\ + \beta_3(\Delta SALES_{it}/ASSETS_{it-1}) + \beta_4(\Delta SALES_{it-1}/ASSETS_{it-1}) + \varepsilon_{it} \quad (2)$$

Where:

PROD	Cost of goods sold + change in inventory.
ASSETS	Total assets.
SALES	Net sales.
ΔSALES	Change in net sales.

Abnormal production costs (*ABNPROD*) are then the difference between actual costs and the 'normal' level of expenses resulting from the use of the estimated coefficients from Eq. (2).

3.2.2. Accrual earnings management proxy

The estimation of our accrual manipulation proxy is based on abnormal working capital accruals (*AWCA*) following the DeFond and Park (2001) methodology. Indeed, as stated in Kim, Chung, and Firth (2003, p. 347), while 'previous research has widely used discretionary accruals under the Jones (1991) model or its other variants, these models have been criticized as the parameter estimates are biased and measurement errors associated therewith could potentially induce erroneous conclusions about the existence of earnings management (e.g. Bernard & Skinner, 1996; Guay, Kothari, & Watts, 1996; Healy, 1996). DeFond and Park (2001) methodology is independent from potential measurement errors associated with the Jones (1991) model parameters.

Accordingly, abnormal working capital accruals (*AWCA*) are estimated as the difference between the current year's realized working capital accruals and the expected level of working capital accruals, using the following formula (3)[8]:

$$AWCA_{it} = WC_{it} - (WC_{it-1}/S_{it-1}) * S_{it} \quad (3)$$

Where:

AWCA	Abnormal working capital accruals.
WC	Operating working capital. It is calculated as current assets after subtracting cash and cash equivalent, less current liabilities net of the current portion of long term debt.
S	Net Sales.

AWCAs calculated above are divided by beginning total assets to adjust for firm size. Signed values of *AWCA* are used as we are interested in observing if discretionary accruals are used to boost or decrease earnings.

3.3. Earnings management and pre-bankruptcy level of financial distress

To analyse the effect of pre-bankruptcy level of financial distress on earnings management tools and gather the evidence to test our hypothesis, we estimate the following model (4):

$$Y_{it} = \alpha + \beta_1 DISTRESS_{it} + \beta_2 SIZE_{it} + \beta_3 CFO_{it} + \beta_4 LEV_{it} + \beta_5 ROA_{it} + \\ + \beta_6 EISSUE_{it} + \beta_7 DISSUE_{it} + \beta_8 GROWTH_{it} + \varepsilon_{it} \quad (4)$$

Where:

Y	*ABNCFO*, *ABNPROD* and *AWCA*.
DISTRESS	Firms' level of pre-bankruptcy financial distress.
SIZE	Natural logarithm of total assets.
CFO	Cash flow from operations divided by total assets.
LEV	Leverage measured as total liabilities over total equity.
ROA	Return on asset calculated as operating profit over total assets.
EISSUE	Annual change in shareholder equity.
DISSUE	Annual change in liabilities.
GROWTH	Annual change in net sales.

In the hypothesis that managers of firms with higher pre-bankruptcy levels of financial distress use real transaction manipulation to hide poor financial performance, then increasing their earnings, more than firms with lower levels of distress, we would expect β_1 to be negative and significant when the dependent variable is *ABNCFO*. Indeed, it would be consistent with an abnormally low level of cash flow, given reported sales, among firms with higher levels of financial distress. It would mean that, among such firms, a lower amount of sales are converted into cash and it can be due to several causes: the concession of abnormal discount to customers, premature sale recognition, more lenient credit terms or, as an extreme scenario, it could potentially be related to a financial statement fraud i.e. recognition of fictitious sales rather than simple earnings management. In the same hypothesis, we would also find β_1 to be significantly positive when the dependent variable is *ABNPROD*. Indeed, it would indicate that firms with higher pre-bankruptcy levels of financial distress exhibit higher production costs as a result of an overproduction which lowers fixed costs per unit with a consequent decrease of the cost of goods sold and an increase of the operating margin (Roychowdhury, 2006). Finally, when the dependent variable is *AWCA*,

[6] The full explanation about the development of Roychowdhury (2006) models can be found in the appendix B of that paper, p. 367–368.

[7] Gunny (2010) affirms that the Roychowdhury (2006) definition of production costs overcomes several disadvantages from using cost of goods sold alone. She explains in her footnote 12 that the use of the above presented definition of production costs as a real earnings manipulation proxy instead of the cost of goods sold expense alone 'mitigate the confounding influence of accrual management. (…) For example, if a manager decided to postpone the write-down of obsolete inventory in an effort to decrease reported cost of goods sold, this action would manifest as abnormally low cost of goods sold expense. Using cost of goods sold as the real earnings manipulation proxy would misclassify accrual management as real manipulation. By examining production costs as sum of cost of goods sold and difference in inventory, the manager's action would not affect production costs because the change in inventories would be correspondingly higher to offset lower cost of goods sold'.

[8] The full explanation of this formula can be found in the appendix of the DeFond and Park (2001) paper, p. 401–402.

if firms with higher pre-bankruptcy levels of financial distress rely less on income-increasing accruals to boost earnings than firms with lower levels of distress, we should observe a negative and significant coefficient β_1.

A set of control variables is also included in the regression to control for other firm-level factors that can influence earnings management. Earlier studies have found that financial leverage (*LEV*) is positively related to earnings management (e.g. Dechow, Sloan, & Sweeney, 1995). Cash flow from operations (*CFO*) and return on assets (*ROA*) are included in the model to control for extreme performance, which may affect the level of earnings management (e.g. Kothari, Leone, & Wasley, 2005). Growth of the firm can affect the extent of earnings management (Carey & Simnett, 2006); for this reason the variables *GROWTH* is introduced. Shan, Taylor, and Walter (2013) show that failure to control for changes in financing can result in significant earnings management measurement errors and erroneous inferences. *DISSUE* and *EISSUE* are included in the model to control for debt and equity issuance, respectively. *SIZE*, measured as the natural logarithm of total assets, is included as earnings management is affected by firm dimension (e.g. Park & Shin, 2004).

All models will be estimated using OLS and include year and industry dummy variables. P-values will be calculated using t-statistics from robust standard errors clustered by firm to correct for serial correlation. All continuous variables have been winsorised at the 1st and the 99th percentile to avoid findings being biased due to the presence of outliers. Finally, a control for the presence of multicolinearity will also be carried out[9].

3.4. Sample selection

To select the sample for this research, we started from the webpage 'www.publicidadconcursal.es'[10]. The use of this webpage was necessary since, in Spain, there is not a commercial database of bankrupt companies while, through this website, it is possible to check whether a company has entered into a legal procedure for bankruptcy, when this happened and the status of the procedure (i.e. liquidation, reorganization, pending decision) at the time of the access. We collected from this website the name and the fiscal number of all companies[11] in a bankruptcy procedure in the twelve Commercial Justice Courts in the region of Madrid from May to June 2010, which totalled 1387 during this period of time[12]. Using the fiscal number of these companies, we searched, manually and one by one, their financial statements on SABI database[13] obtaining the required data for 914 firms. We aim to analyse earnings management practices between the first and the third year before the beginning of the legal procedure for bankruptcy because this is the period when extant evidence indicates that bankrupt companies manipulate earnings more than other entities (e.g. Campa & Camacho-Miñano, 2014; García Lara et al., 2009)[14]. For this reason, we excluded all firms without an adequate time series. At the end of this process, the sample is composed of 362 unique bankrupt firms over a three-year period for a total of 1086 firm-year observations.

[9] A test of multicollinearity includes the analysis of the VIF (Variance Inflation Factor) test, ensuring that the average of the model is below the critical value of 10 as indicated by Kennedy (2008).

[10] According to article 198 of the Spanish Act on Insolvency, this website is 'accessible free of charge on the Internet and shall publish all the insolvency resolutions that require publication pursuant to the provisions of this Act'.

[11] Individual firms have been excluded from the analysis. In our initial sample there was only one in the bankruptcy webpage during the period of sample extraction.

[12] The region of Madrid is one of the most important in Spain and, every year, it includes around 19% of all bankrupt firms in the whole of Spain (Source: INE – Spanish National Statistics Institute). The choice of the time series is related to the time when the data collection happened and also reflects a feasibility analysis since data have been manually collected.

[13] SABI is a database provided by Bureau Van Dijk that includes comprehensive information of more than 1,250,000 companies operating in Spain.

[14] Our analysis covers the years from 2006 to 2009. As the economic crisis the EU is facing at the moment this paper is prepared started at the end of 2008, it is unlikely that the overall findings and conclusions are significantly affected by this event.

Table 2
Descriptive analysis.

	N.	Mean	Median	St. Dev	1st Quartile	3rd Quartile
ABNCFO	1086	0.077	0.065	0.507	−0.085	0.281
ABNPROD	1086	0.049	−0.009	0.223	−0.065	0.084
AWCA	1086	−0.251	0.000	2.039	−0.166	0.163
DISTRESS	1086	0.444	0.000	0.497	0.000	1.000
SIZE	1086	14.776	13.598	1.646	13.598	15.887
CFO	1086	−0.429	0.001	8.296	−0.159	0.074
LEV	1086	8.963	3.960	25.470	0.975	9.502
ROA	1086	−0.078	0.020	0.313	−0.079	0.052
EISSUE	1086	−0.328	0.380	2.513	−0.363	0.219
DISSUE	1086	0.312	0.107	0.919	−0.080	0.368
GROWTH	1086	0.678	−0.005	4.291	−0.194	0.238

ABNCFO is a proxy for sales manipulation. *ABNPROD* is a proxy for production cost manipulation. *AWCA* is the signed value of the abnormal working capital accrual (DeFond & Park, 2001) scaled by beginning total assets. *DISTRESS* is a dummy variable which takes 1 for firms with sign of financial distress in accordance with the methodology developed by Fich and Slezak (2008). *SIZE* is the natural logarithm of the total assets. *CFO* is cash flow from operations divided by total assets. *LEV* is calculated as total liabilities over total equity. *ROA* is operating profit divided by total assets. *EISSUE* is the annual change in shareholder equity. *DISSUE* is annual change in total liabilities. *GROWTH* is the annual change in net sales.

4. Results and discussion

4.1. Descriptive statistics and univariate analyses

Table 2 shows descriptive statistics for the variables used in our study.

Table 2 indicates that 44.4% of our sample shows signs of financial distress in accordance with the definition of distress used. Descriptive statistics also indicate that *ABNCFO* and *ABNPROD* measure 7.7% and 4.9% of company assets respectively while discretionary accruals are, on average, negative and about 25% of total assets. The variables *ROA* and *CFO* show negative averages, perfectly in line with the characteristics of the firms included in the sample. Firms also present high levels of debt as highlighted by the leverage ratio which shows a mean of 8.96. The average total assets are 2.6 millions, in line with what the European Union defines as 'small companies'[15].

Pearson correlation matrix is reported in Table 3.

Table 3 exhibits a non significant correlation between *ABNCFO* and firms with a higher pre-bankruptcy level of financial distress ($\rho = 0.010$; p-value = 0.718). A positive and significant correlation between firms with higher pre-bankruptcy levels of financial distress and *ABNPROD* ($\rho = 0.479$; p-value = 0.000) suggests that these entities exhibit higher production costs than companies with lower levels of distress. Finally, discretionary accruals are not significantly correlated with the variable *DISTRESS* ($\rho = 0.005$; p-value = 0.858).

It is worth noticing that all the above mentioned variables have significant correlations with other factors that will be used as control variables in the regression models, therefore only a multivariate analysis can provide statistically reliable evidence about the hypothesis proposed.

4.2. Regression analysis

Table 4 reports the estimation of our model (4) which investigates earnings management practices in relation to the pre-bankruptcy level of companies' financial distress.

The model is always robust (p-value = 0.000 regardless of the dependent variable used) and the R^2 goes from 29.1% (column A) to 86.8% (column C). β_1 from column A is negative and significant at the 10% level ($\beta = -0.049$; p-value = 0.100), evidence that companies

[15] A small enterprise is defined as an enterprise which employs fewer than 50 persons and whose annual turnover and/or annual balance sheet total does not exceed EUR 10 million (Commission Recommendation 2003/361/EC of 6 May 2003).

Table 3
Pearson correlation matrix.

	ABNCFO	ABNPROD	AWCA	DISTRESS	SIZE	CFO	LEV	ROA	EISSUE	DISSUE	GROWTH
ABNCFO	1										
ABNPROD	−0.078**	1									
AWCA	−0.185***	−0.039	1								
DISTRESS	0.010	0.479***	0.005	1							
SIZE	−0.060**	−0.229***	−0.068**	−0.089***	1						
CFO	0.271***	−0.103***	−0.073**	−0.054**	−0.089***	1					
LEV	−0.026	−0.055*	0.015	−0.124***	0.026	0.081***	1				
ROA	−0.005	−0.810***	0.075**	−0.542***	0.278***	−0.007	0.104***	1			
EISSUE	−0.080***	−0.228***	−0.044	−0.263***	0.075**	0.083***	0.054*	0.229***	1		
DISSUE	−0.274***	0.156***	−0.084***	−0.061**	0.145***	−0.274***	0.018	−0.022	−0.025	1	
GROWTH	−0.005	−0.025	−0.919***	−0.092***	0.131***	0.004	0.004	0.026	0.093***	0.208***	1

Notes: *, **, *** indicate that a coefficient is statistically significant at the 10%, 5%, and 1% level or better.
ABNCFO is a proxy for sales manipulation. *ABNPROD* is a proxy for production cost manipulation. *AWCA* is winsorised value of the abnormal working capital accruals as defined by DeFond and Park (2001). *DISTRESS* is a dummy variable which takes 1 for firms with sign of financial distress in accordance with the methodology developed by Fich and Slezak (2008). *SIZE* is natural logarithm of total assets. *CFO* is cash flow from operations divided by total assets. *LEV* is leverage measured as total liabilities over total equity. *ROA* is return on asset calculated as operating profit over total assets. *EISSUE* is annual change in shareholder equity. *DISSUE* is annual change in liabilities. *GROWTH* is annual change in net sales.

with higher pre-bankruptcy levels of financial distress exhibit lower inflows of cash than other firms, given similar sales. It could be interpreted as presence of sale manipulation (e.g. credit term extension, fictitious sales, etc.) aimed at inflating earnings or, given the characteristics of our sample, the results of unusual discounts that would, at the same time, increase the level of sales and accelerate the availability of (abnormally lower) cash.

Table 4
Earnings manipulation and financial distress.

	(A)	(B)	(C)
Dependent variable	ABNCFO	ABNPROD	AWCA
INTERCEPT	0.098	0.204***	−0.408*
	(0.56)	(3.84)	(−1.87)
DISTRESS	**−0.049***	**0.149*****	**−0.087***
	(−1.65)	**(9.15)**	**(−1.75)**
SIZE	−0.004	−0.018***	0.024*
	(−0.34)	(−5.17)	(1.67)
CFO	0.014***	−0.001	−0.109**
	(3.02)	(−1.51)	(−2.08)
LEV	−0.000	−0.000	0.000
	(−0.91)	(−0.47)	(1.18)
ROA	0.035	−0.143***	0.204***
	(1.32)	(−3.64)	(3.14)
EISSUE	−0.025***	0.007*	0.025***
	(−3.43)	(−1.92)	(2.69)
DISSUE	−0.141***	0.040***	0.211***
	(-4.16)	(4.57)	(3.49)
GROWTH	0.005	−0.001	−0.450***
	(0.76)	(−0.64)	(−29.08)
Observations	1086	1086	1086
R-squared	0.291	0.482	0.869
F-Stat	20.18***	38.69***	86.04***
Year and Industry dummies	Yes	Yes	Yes
VIF of the model	1.30	1.30	1.30

Notes: T-statistics (in parentheses below the coefficients) are calculated using standard errors clustered by firm. For clarity, the year-specific and industry-specific intercepts are omitted. *, **, *** indicate that a coefficient is statistically significant at the 10%, 5%, and 1% level or better.
Regression models: $Y_{it} = \alpha + \beta_1 DISTRESS_{it} + \beta_2 SIZE_{it} + \beta_3 CFO_{it} + \beta_4 LEV_{it} + \beta_5 ROA_{it} + \beta_6 EISSUE_{it} + \beta_7 DISSUE_{it} + \beta_8 GROWTH_{it} + \varepsilon_{it}$
$Y_{it} = ABNCFO$ (Col. A), *ABNPROD* (Col. B), *AWCA* (col. C)
ABNCFO is a proxy for sales manipulation. *ABNPROD* is a proxy for production cost manipulation. *AWCA* is winsorised value of the abnormal working capital accruals as defined by DeFond and Park (2001). *DISTRESS* is a dummy variable which takes 1 for firms with sign of financial distress in accordance with the methodology developed by Fich and Slezak (2008). *SIZE* is the natural logarithm of the total assets. *CFO* is cash flow from operations divided by total assets. *LEV* is leverage measured as total liabilities over total equity. *ROA* is return on asset calculated as operating profit over total assets. *EISSUE* is annual change in shareholder equity. *DISSUE* is annual change in liabilities. *GROWTH* is annual change in net sales.

The same coefficient β_1, in column B, is positive and significant at the 1% level ($\beta = 0.149$; p-value $= 0.000$). It suggests that firms which observe severe pre-bankruptcy financial distress exhibit higher production costs than other bankrupt companies. It might be the result of an increase in the level of production which lowers fixed costs per unit or alters the changes in inventories with a consequent decrease of the cost of goods sold and an increase of the operating margin (Roychowdhury, 2006).

Column C investigates earnings management through the use of discretionary accruals. β_1 is negative and significant at the 10% level ($\beta = -0.087$; p-value $= 0.081$). It indicates that firms with lower pre-bankruptcy levels of financial distress exhibit more income-increase abnormal accruals in comparison with highly distressed bankrupt companies.

At this stage one could argue that estimating the three models presented in Table 4 separately does not take into account the fact that companies could use sale, production cost and accrual manipulation, to a certain extent, at the same time. To deal with this issue, in line with Liu, Hodgkinson, and Chuang (2014), we estimate panel system equations by considering the simultaneity of real activity and accrual manipulation. Systems are estimated by taking into account the interdependences among the equations in the system (Wooldridge, 2002) [16]. In our case, model (4) including our proxies for sale, production cost and accrual manipulation is estimated simultaneously using the generalized method of moments (GMM). Indeed, this simultaneous approach takes account of correlations in the residuals across equations and allows us to consider the interdependence of sale, cost and accrual manipulation activities. Such an empirical approach enables us to treat real activity and accrual manipulation as interdependent and these types of earnings manipulation as not mutually exclusive (Liu et al., 2014). Results are presented in Table 5.

The sign of β_1 is, in all columns, in line with Table 4. In addition, using GMM, the significance of the coefficients related to the use of sales and accrual manipulation becomes stronger as indicated by the higher values (in absolute terms) of their *t*-stat. The evidence that suggests that firms with higher pre-bankruptcy levels of financial distress exhibit, before bankruptcy, more income-increasing manipulation through real activity management and less through the use of discretionary accruals in comparison with companies with lower pre-bankruptcy levels of distress is also confirmed when these earnings management tools are considered as interdependent and not mutually exclusive.

One reason for these results could be that less distressed firms need less 'pervasive/forceful' earnings management tools. Managers of less financially distressed firms have weaker incentives to manage earnings

[16] A system is a group of equations containing unknown parameters.

Table 5
Earnings manipulation and financial distress using generalized method of moments (GMM).

	(A)	(B)	(C)
Dependent variable	ABNCFO	ABNPROD	AWCA
INTERCEPT	−0.385***	0.065*	0.208
	(−4.45)	(1.76)	(1.28)
DISTRESS	**−0.048****	**0.036*****	**−0.081****
	(**−2.37**)	(**4.18**)	(**−2.10**)
SIZE	0.034***	−0.008***	−0.010
	(6.06)	(−3.51)	(−0.92)
CFO	1.116***	−0.10	−1.016***
	(47.30)	(−1.01)	(−22.75)
LEV	−0.000	0.000	0.000
	(−0.05)	(−1.32)	(0.04)
ROA	−0.099**	−0.656***	0.804***
	(−2.45)	(−38.74)	(10.72)
EISSUE	0.001	0.000	−0.003
	(0.22)	(0.01)	(−0.37)
DISSUE	−0.025**	0.044***	0.086***
	(−1.97)	(8.04)	(3.54)
GROWTH	0.008*	0.001	−0.485***
	(1.86)	(0.29)	(−57.01)
Observations	1086	1086	1086
'R-squared'	0.713	0.723	0.789
Chi2	2649.51***	2997.45***	3989.52***
Year and Industry dummies	Yes	Yes	Yes
VIF of the model	N/A	N/A	N/A

Notes: T-statistics (in parentheses below the coefficients) are calculated using standard errors clustered by firm. For clarity, the year-specific and industry-specific intercepts are omitted. *, **, *** indicate that a coefficient is statistically significant at the 10%, 5%, and 1% level or better.
Regression models: $Y_{it} = \alpha + \beta_1 DISTRESS_{it} + \beta_2 SIZE_{it} + \beta_3 CFO_{it} + \beta_4 LEV_{it} + \beta_5 ROA_{it} + \beta_6 EISSUE_{it} + \beta_7 DISSUE_{it} + \beta_8 GROWTH_{it} + \varepsilon_{it}$
$Y_{it} = ABNCFO$ (Col. A), *ABNPROD* (Col. B), *AWCA* (col. C)
ABNCFO is a proxy for sales manipulation. *ABNPROD* is a proxy for production cost manipulation. *AWCA* is winsorised value of the abnormal working capital accruals as defined by DeFond and Park (2001). *DISTRESS* is a dummy variable which takes 1 for firms with sign of financial distress in accordance with the methodology developed by Fich and Slezak (2008). *SIZE* is the natural logarithm of the total assets. *CFO* is cash flow from operations divided by total assets. *LEV* is leverage measured as total liabilities over total equity. *ROA* is return on asset calculated as operating profit over total assets. *EISSUE* is annual change in shareholder equity. *DISSUE* is annual change in liabilities. *GROWTH* is annual change in net sales.

opportunistically, according to Kim and Sohn (2013). These authors also give evidence of the use of real activity manipulation for the achievement of short-term targets (i.e. the attempt to avoid bankruptcy in our study), even if it is more costly and has 'adverse consequences on long-term firm value' (Kim & Sohn, 2013, p. 21). Recently, Trombetta and Imperatore (2014) argue that real earnings manipulation decreases when the intensity of the crisis is low while it increases when the crisis is acute. Our results endorse the same idea but measuring the crisis of a firm with a proxy of financial distress index. In addition, other studies produce evidence that firms prefer real activity manipulation because it damages investor trust less than accruals (Hewitt, Hodge, & Pratt, 2014). In a pre-bankruptcy situation, financially distressed companies need the trust of several stakeholders such as creditors, judges, and insolvency administrators, so this could be another explanation of the switch between earnings management tools. Our findings also suggest that the use of real activity manipulation depends more on the circumstances that can encourage firms to manipulate earnings (high levels of distress that can increase the probability of filing for bankruptcy) than the cost of this tool for the firm (Kim & Sohn, 2013). Indeed, although real activity manipulation has worse consequences on performance (Gunny, 2010) and on long-term profitability and competitiveness (Kothari et al., in press) than accruals, it is used as the 'last chance' to avoid bankruptcy.

We note that our evidence is in contrast with that of Zang (2012) that indicates that healthier firms use more real activity and less accrual based earnings management. One explanation could be the relevance of financial distress and the size of firms. Our sample includes small companies and financial distress is non-temporary so the pressure it puts on managers is different from the effect it has on other companies. Our results also go in the opposite direction to those of Enomoto et al. (in press) and Geiger and van der Laan Smith (2010) that show that real activity management is preferred to accrual manipulation in countries with stronger investor protection. Our sample is from a code law country (Spain) with weak investor protection (Leuz et al., 2003) so our findings indicate that the financial context of firms conditions the way earnings are manipulated more than the institutional setting in which firms operate. If the level of financial distress is significantly high, companies increase earnings by focusing more on managing real transactions and rely less on managing accruals. Our conclusion that the level of pressure on managers affects earnings management behaviours more than the institutional settings and the status of company (large/listed versus small/unlisted) is also supported by the fact that our results are in line with other studies such as Kim et al. (2013), Chamberlain et al. (2014) and Trombetta and Imperatore (2014) that use data from a common law context (the United States) and listed companies.

4.3. Robustness tests

To increase the reliability of the findings presented above, some robustness tests have been carried out. We repeated our regression analysis using three other estimations of financial distress.

The first employs the methodology implemented by Carcello and Neal (2000). It is applicable to our sample since it is also exclusively based on accounting data and, furthermore, this methodology has been already applied in code law contexts, including Spain as well (e.g. Arnedo, Lizarraga, & Sanchez, 2009; Arnedo, Lizarraga, & Sánchez, 2008; Tykvová & Borell, 2012). Carcello and Neal (2000) estimate the probability of failure of a company on the basis of the following formula (5):

$$\int_{-\infty}^{y} \frac{1}{\sqrt{2\pi}} e^{\frac{z^2}{2}} dz \tag{5}$$

Y is calculated based upon the following Eq. (6) from Zmijewski (1984):

$$Y = -4.336 - 4.513 * ROA + 5.679 * FNL - 0.004 * LIQ \tag{6}$$

Where:

ROA Net income divided by total assets.
FNL Total debt divided by total assets.
LIQ Current assets divided by current liabilities.

We then create a dummy variable, *DISTRESS*, to identify firms that are financially distressed. It takes 1 if the probability of failure calculated above exceeds 28% and 0 otherwise, in accordance with previous studies (e.g. Davis & Ashton, 2002; Krishnan & Krishnan, 1997). However, since extant evidence indicated that bankrupt companies manipulate earnings upwards when close to bankruptcy (e.g. García Lara et al., 2009), the use of the ROA in the formula above could affect the classification of a firm as being (or not) financially distressed. In such a scenario, upward earnings management would opportunistically decrease the probability of failure. To deal with such an issue, we also carry out a sensitivity analysis on the variable *DISTRESS* by estimating it using lower probabilities of failure for the identification of financially distressed companies. More precisely, we will also use a 25% and a 20% probability of failure to classify a firm as financially distressed.

The second distress indicator is based on the continuous Z-score in accordance with the probability of failure developed by Zmijewski (1984)

Table 6
Robustness test: earnings manipulation and other estimations of financial distress.

PANEL A – Carcello and Neal (2000) measure of financial distress

	(A)	(B)	(C)
Dependent variable	ABNCFO	ABNPROD	AWCA
INTERCEPT	0.145	0.223***	−0.236
	(0.70)	(3.53)	(−1.03)
DISTRESS	**−0.134*** **	**0.067*** **	**−0.108***
	(−3.16)	**(3.07)**	**(−1.94)**
SIZE	−0.003	−0.022***	0.029*
	(−0.27)	(−4.97)	(1.88)
CFO	0.014***	−0.002**	−0.011**
	(3.02)	(−1.98)	(−2.05)
LEV	−0.000	−0.000	0.001*
	(−0.32)	(−0.83)	(1.88)
ROA	0.047***	−0.116***	0.149***
	(2.90)	(−3.47)	(2.94)
EISSUE	−0.024***	−0.011***	0.027***
	(−3.45)	(−3.04)	(2.89)
DISSUE	−0.135***	0.033***	0.221***
	(−4.05)	(3.80)	(3.65)
GROWTH	0.004	−0.000	−0.451***
	(0.68)	(−0.18)	(−29.11)
Observations	1086	1086	1086
R-squared	0.303	0.373	0.868
F-Stat	21.31***	18.46***	89.14***
Year and Industry dummies	Yes	Yes	Yes
VIF of the model	1.34	1.34	1.34

PANEL B – Zmijewski (1984) measure of financial distress

	(A)	(B)	(C)
Dependent variable	**ABNCFO**	**ABNPROD**	**AWCA**
INTERCEPT	0.103	0.169***	−0.141
	(0.64)	(2.90)	(−0.68)
DISTRESS	**−0.110***	**0.017*** **	**−0.040*** **
	(−1.84)	**(4.13)**	**(−3.27)**
SIZE	−0.010	−0.014***	0.014
	(−0.89)	(−3.54)	(1.07)
CFO	0.014***	−0.002*	−0.011*
	(3.07)	(−1.66)	(−1.91)
LEV	−0.000	−0.000	0.000
	(−0.91)	(−0.07)	(1.44)
ROA	0.021	−0.070**	0.038
	(0.72)	(−2.38)	(0.75)
EISSUE	−0.024***	−0.024***	0.024***
	(−3.28)	(−3.28)	(2.59)
DISSUE	−0.137***	0.030***	0.241***
	(−4.06)	(3.38)	(3.93)
GROWTH	0.005	−0.000	−0.459***
	(0.64)	(−0.03)	(−25.92)
Observations	1086	1086	1086
R-squared	0.294	0.421	0.859
F-Stat	19.49***	18.95***	72.77***
Year and Industry dummies	Yes	Yes	Yes
VIF of the model	1.46	1.46	1.46

PANEL C – Altman's (2000) Z-Score measure of financial distress

	(A)	(B)	(C)
Dependent variable	**ABNCFO**	**ABNPROD**	**AWCA**
INTERCEPT	−0.398***	0.032	−0.069
	(5.55)	(0.70)	(−0.44)
DISTRESS	**0.011***	**0.006**	**0.030*** **
	(1.88)	**(1.21)**	**(3.92)**
SIZE	0.029***	−0.006**	−0.003
	(6.20)	(−2.14)	(−0.27)
CFO	1.125***	−0.015	−0.999***
	(37.93)	(−0.93)	(−16.64)
LEV	0.000	0.000	0.000
	(1.40)	(1.10)	(0.26)
ROA	0.011	−0.749***	0.603***
	(0.24)	(−17.59)	(7.45)
EISSUE	−0.003	−0.001	0.001
	(−0.67)	(−0.38)	(0.08)
DISSUE	−0.051***	0.044***	0.097**
	(−3.50)	(5.27)	(2.28)

Table 6 (*continued*)

PANEL C – Altman's (2000) Z-Score measure of financial distress

	(A)	(B)	(C)
Dependent variable	**ABNCFO**	**ABNPROD**	**AWCA**
GROWTH	0.005	0.000	−0.486***
	(1.22)	(0.09)	(−18.32)
Observations	1086	1086	1086
R-squared	0.878	0.732	0.795
F-Stat	243.00***	64.81***	129.66***
Year and Industry dummies	Yes	Yes	Yes
VIF of the model	1.45	1.45	1.45

Notes: T-statistics (in parentheses below the coefficients) are calculated using standard errors clustered by firm. For clarity, the year-specific and industry-specific intercepts are omitted. *, **, *** indicate that a coefficient is statistically significant at the 10%, 5%, and 1% level or better.
Regression models: $Y_{it} = \alpha + \beta_1 DISTRESS_{it} + \beta_2 SIZE_{it} + \beta_3 CFO_{it} + \beta_4 LEV_{it} + \beta_5 ROA_{it} + \beta_6 EISSUE_{it} + \beta_7 DISSUE_{it} + \beta_8 GROWTH_{it} + \varepsilon_{it}$
$Y_{it} = ABNCFO$ (Col. A), *ABNPROD* (Col. B), *AWCA* (col. C)
ABNCFO is a proxy for sales manipulation. *ABNPROD* is a proxy for production cost manipulation. *AWCA* is winsorised value of the abnormal working capital accruals as defined by DeFond and Park (2001). *DISTRESS* is an indicator of the probability of bankruptcy measured in accordance with the Altman Z-Score methodology for unlisted companies. *SIZE* is the natural logarithm of the total assets. *CFO* is cash flow from operations divided by total assets. *LEV* is leverage measured as total liabilities over total equity. *ROA* is return on asset calculated as operating profit over total assets. *EISSUE* is annual change in shareholder equity. *DISSUE* is annual change in liabilities. *GROWTH* is annual change in net sales.

and described in the Eq. (6) above. Higher value of the Z-score indicates higher levels of financial distress.

Finally, despite the criticism raised by García Lara et al. (2009) in relation to the use of the Altman's (1968) Z-score and highlighted in our endnote 2, our third measure of financial distress follows the revised Altman's (2000) Z-score which is applicable to unlisted companies. The Z-score for private firms, in accordance with Altman (2000), is estimated on the basis of the following formula (7):

$$Z = 0.717*X1 + 0.847*X2 + 3.107*X3 + 0.420*X4 + 0.998*X5 \qquad (7)$$

Where:

X1 Current assets less current liabilities divided by total assets.
X2 Retained earnings divided by total assets.
X3 Earnings before interest and taxes divided by total assets.
X4 Book value of equity divided by total liabilities.
X5 Sales divided by total assets.

On the contrary of the Zmijewski's (1984) approach, in this case higher values of the Z-score indicate lower probability of financial distress.

The estimation of models (4) using these alternative measures of distress is reported in Table 6.

Panel A of Table 6 reports results for our two proxies of real earnings manipulation and our estimation of discretionary accruals using the Carcello and Neal (2000) index of distress. The evidence is in line with that of Table 4. β_1 from column A is negative and significant at the 1% level ($\beta = -0.134$; p-value = 0.002), evidence that companies with higher pre-bankruptcy levels of financial distress exhibit lower inflows of cash than other firms, given similar sales. The same coefficient β_1, in column B, is positive and significant at the 1% level ($\beta = 0.067$; p-value = 0.002). Finally, β_1 is negative and significant at the 10% level ($\beta = -0.108$; p-value = 0.054) in column C. These findings are confirmed also when we use the 25% and 20% probability of failure to identify firms that are financially distressed (untabled)[17].

[17] The coefficients associated with the real earnings management proxies always remain significant at the 1% level. The significance of the coefficient related to the abnormal accruals is significant at the 10% level when the 25% threshold is applied while it is significant, only if considered one-tail, when the 20% threshold is employed.

Consistent results can be observed in Panel B of Table 6 that uses the Zmijewski (1984) index. It shows a negative and significant β_1 when the dependent variable is *ABNCFO* (column A) and *AWCA* (column C) while β_1 is positive and significant when the dependent variable is *ABNPROD* (column B).

Finally, when the Altman's (2000) Z-score is used, results are supported for sales and accrual manipulation. The sign of these coefficients goes in the opposite direction than the other panels because, in this case, higher values of the score mean lower level of distress. The coefficient β_1 for production cost manipulation (column B) becomes instead not significant.

Overall, all this evidence supports our main conclusion that firms with higher pre-bankruptcy levels of financial distress increase earnings through real transaction manipulation while higher income-increasing discretionary accruals can be observed among firms with lower pre-bankruptcy levels of financial distress[18].

Despite the limitations of the Jones (1991) type models for the estimation of discretionary accruals highlighted in the methodology section, we use one of those as a robustness test. More precisely, we use the definition of discretionary accruals developed by Kasznik (1999), already employed in studies focused on distressed firms (e.g. García Lara et al., 2009). These abnormal working capital accruals are obtained by estimating the following regression (8) for all firms in industry i for year t:

$$WCA_{it}/ASSETS_{it-1} = \alpha + \beta_1(1/ASSETS_{it-1}) + \beta_2(\Delta SALES_{it}/ASSETS_{it-1}) + \\ + \beta_3(\Delta CFO_{it}/ASSETS_{it-1}) + \varepsilon_{it} \quad (8)$$

Where:

WCA	Working capital accruals (calculated as current assets net of cash and short-term investments minus current liabilities net of short-term debts);
ASSETS	Total assets;
ΔSALES	Change in net sales;
ΔCFO	Change in cash flow from operations.

Abnormal accruals are then defined as the residuals from the regression above. They are included as dependent variable in our model (4) using both our main estimation of financial distress (i.e. Fich & Slezak, 2008) and those employed in our robustness tests (i.e. Altman, 2000; Carcello & Neal, 2000 and Zmijewski, 1984). Results are reported in Table 7.

Evidence supports our conclusion that income-increasing discretionary accruals are used more by firms with lower levels of distress, except when the Fich and Slezak (2008) index is used. Indeed, in all the other cases, the coefficient associated with the variable *DISTRESS* is significant at the 10% level when the Carcello and Neal (2000) methodology is used and at the 1% level when the Zmijewski (1984) and the Altman's (2000) Z-scores are employed.

Finally, since the indicators of distress are built using financial data that, in turn, could be impacted by earnings manipulation, the findings presented could be affected by endogeneity issues. To deal with such an issue, we re-estimated our original model (4) using a two-stage least square approach. In the first stage, we use our proxy of distress as instrumented variable and each earnings manipulation variable, in turn, together with our measures of profitability, size and growth as instruments. Results (untabled) supports the findings reported in Table 4[19].

[18] GMM have also been used with this different estimations of distress and results (untabled) which are consistent with those provided by OLS.

[19] When *ABNCFO* is the dependent variable $\beta_1 = -0.283$; p-value = 0.000; when *ABNPROD* is the dependent variable $\beta_1 = 0.809$; p-value = 0.000; when *AWCA* is the dependent variable $4 = -0.505$; p-value = 0.000.

Table 7
Robustness tests: alternative measure of discretionary accruals.

	(A)	(B)	(C)	(D)
Dependent variable	DiscAcc	DiscAcc	DiscAcc	DiscAcc
(Distress index)	(Fich & Slezak, 2008)	(Carcello & Neal, 2000)	(Zmijewski, 1984)	(Altman's Z-score)
INTERCEPT	−1.288***	−1.329***	−1.04***	−0.972***
	(−1.91)	(−5.07)	(−4.37)	(−5.75)
DISTRESS	**−0.038**	**−0.075***	**−0.050*****	**0.029*****
	(**−0.87**)	(**−1.42**)	(**−4.61**)	(**3.59**)
SIZE	0.107***	0.112***	0.092***	0.084***
	(6.33)	(6.62)	(5.92)	(7.80)
CFO	0.003	0.003	0.003	−0.652***
	(0.60)	(0.61)	(0.55)	(−11.74)
LEV	−0.001	−0.000	−0.000	−0.000
	(−0.39)	(−0.07)	(−0.48)	(−0.30)
ROA	0.191***	0.124***	−0.016	0.462***
	(4.82)	(2.82)	(−0.40)	(4.95)
EISSUE	0.018**	0.019**	0.014*	0.004
	(2.20)	(2.40)	(1.79)	(0.45)
DISSUE	0.029	0.034	0.048**	−0.052**
	(1.11)	(1.26)	(1.96)	(−1.98)
GROWTH	0.010**	0.009**	0.007	0.024**
	(2.18)	(1.97)	(1.24)	(2.51)
Observations	1086	1086	1086	1086
R-squared	0.259	0.244	0.294	0.415
F-Stat	12.72***	11.62***	26.06***	40.74***
Year and Industry dummies	Yes	Yes	Yes	Yes
VIF of the model	1.30	1.34	1.47	1.45

NotesN T-statistics (in parentheses below the coefficients) are calculated using standard errors clustered by firm. For clarity, the year-specific and industry-specific intercepts are omitted. *, **, *** indicate that a coefficient is statistically significant at the 10%, 5%, and 1% level or better.
Regression model: $DiscAcc_{it} = \alpha + \beta_1 DISTRESS_{it} + \beta_2 SIZE_{it} + \beta_3 CFO_{it} + \beta_4 LEV_{it} + \beta_5 ROA_{it} + \beta_6 EISSUE_{it} + \beta_7 DISSUE_{it} + \beta_8 GROWTH_{it} + \varepsilon_{it}$
DiscAcc is the winsorised value of the discretionary accruals as defined by Kasznik (1999). *DISTRESS* is a dummy variable which takes 1 for firms with sign of financial distress in accordance with the methodology developed by Fich and Slezak (2008) – Column A; a dummy variable which takes 1 for firms with sign of financial distress in accordance with the methodology developed by Carcello and Neal (2000) – Column B; a measure of financial distress in accordance with the methodology developed by Zmijewski (1984) – Column C; an indicator of the probability of bankruptcy measured in accordance with the Altman Z-Score for unlisted companies – Column D. *CFO* is cash flow from operations divided by total assets. *LEV* is leverage measured as total liabilities over total equity. *SIZE* is natural logarithm of total assets. *ROA* is return on asset calculated as operating profit over total assets. *EISSUE* is annual change in shareholder equity. *DISSUE* is annual change in liabilities. *GROWTH* is annual change in net sales.

5. Conclusions

Extant literature is quite consistent in highlighting that companies going through a bankruptcy procedure make an attempt, before their failure, to increase earnings opportunistically using both real activity and accrual manipulation. This paper investigates whether the use of these two earnings management tools (real activities and accruals) is related to the different pre-bankruptcy levels of financial distress of firms that file for bankruptcy. The main conclusion is that the level of financial distress does affect the way earnings are manipulated. The more serious the pre-bankruptcy levels of distress, the more managers manipulate earnings through real transactions and the less through discretionary accruals. It would suggest that in the presence of more complex financial situations, managers try to inflate earnings using the less detectable and more effective strategy in order to make an attempt aimed at avoiding the several negative aspects that this situation can cause, that is an involvement in a bankruptcy procedure with all its consequences and implications such as the potential damages to shareholders, investors, employees and social wellbeing.

Our evidence may have several implications for regulators and practitioners. It suggests that, at least in the presence of a legal procedure

for bankruptcy, the choice of one earnings management tool over another can be driven by firms' pre-bankruptcy level of financial distress. It indicates to regulators that the outcome of the bankruptcy law in a country could provide an indicator about the level of pressure on management among firms approaching bankruptcy, which, in turn, affects the earnings management tools thus conditioning the areas of an entity that need closer attention. Our findings also provide auditors with an efficient technique to help them in making going concern decisions because they can concentrate their efforts on analysing real operation manipulation when companies are in severe financial distress, although these practices are more difficult to detect and easier to camouflage (Kothari et al., in press). Moreover, investors should be cautious about earnings information (Enomoto et al., in press) especially when firms are in financial distress. Analysts should also consider the effect of tighter accounting rules on earnings management, taking into account the financial situation of firms as exceptional circumstances that can skew the implementation of some accounting standards. Our evidence is in line with other studies on manager pressure and earnings manipulation tools focused on common law countries. This would indicate that the conditions under which managers work impact on the way earnings are manipulated more than the institutional settings in which firms operate.

This paper is not free from limitations and can give avenues for future research. The inference is based on the analysis of only one region of a country and on companies filing for bankruptcy in one particular year, so more research is needed in order to generalize our results. A significant number of companies have been deleted from our initial sample because they do not make their financial statement public. Additionally, Beneish (1997, p. 271) indicates that extreme financial distress situations 'limit the usefulness of accrual models'. This might also have had an effect on our results. Finally, we are aware that there are also contexts where the bankruptcy process is a way to deal with companies' financial problems and to take actions (e.g. possible mergers, acquisitions, etc.) that ensure their future viability. Our findings might not hold in such scenarios since the level of pressure that a potential bankruptcy procedure creates on management would be less significant.

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References

Achleitner, A. K., Guenther, N., Kaserer, C., & Siciliano, G. (2014). Real earnings management and accrual-based earnings management in family firms. *The European Accounting Review*, *23*(3), 431–461, http://dx.doi.org/10.1080/09638180.2014.895620.

Adu-Boateng, D. (2011). A theoretical construct for explaining the impact of financial distress on unethical earnings management decisions. *Journal of American Academy of Business*, *16*(2), 89–95.

Alhadab, M., Clacher, I., & Keasey, K. (2013a). Real and accruals earnings management around initial public offering under different regulatory environments. *Working paper*, http://dx.doi.org/10.2139/ssrn.2186502 (Accessed 8 June 2015).

Alhadab, M., Clacher, I., & Keasey, K. (2013b). Real and accrual earnings management and IPO failure risk. *Working paper*, http://dx.doi.org/10.2139/ssrn.2225411 (Accessed 16 June 2015).

Altman, E. I. (1968). Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *Journal of Finance*, *23*(4), 589–609, http://dx.doi.org/10.1111/j.1540-6261.1968.tb00843.x.

Altman, E. I. (1993). *N* (2nd ed.). New York: John Wiley and Sons.

Altman, E. I. (2000). *Predicting financial distress of companies: revisiting the Z-score and ZETA models.* New York University: Stern School of Business (http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.25.1884&rep=rep1&type=pdf Accessed 22 May 2015).

Arnedo, L., Lizarraga, F., & Sánchez, S. (2007). Does public/private status affect the level of earnings management in code-law contexts outside the United States? A study based on the Spanish case. *The International Journal of Accounting*, *42*(3), 305–328, http://dx.doi.org/10.1016/j.intacc.2007.06.004.

Arnedo, L., Lizarraga, F., & Sánchez, S. (2008). Going-concern uncertainties in Pre-bankrupt audit reports: New evidence regarding discretionary accruals and wording ambiguity. *International Journal of Auditing*, *12*(1), 25–44, http://dx.doi.org/10.1111/j.1099-1123.2008.00368.x.

Arnedo, L., Lizarraga, F., & Sanchez, S. (2009). Are Spanish auditors effective in transmitting manipulation practices? An analysis of the ambiguity implicit in modified reports. *International Journal of Accounting, Auditing and Performance Evaluation*, *5*(3), 229–251, http://dx.doi.org/10.1504/IJAAPE.2009.026627.

Baber, W. R., Fairfield, P. M., & Haggard, J. A. (1991). The effect of concern about reported income on discretionary spending decisions: The case of research and development. *The Accounting Review*, *66*(4), 818–829.

Badertscher, B. A. (2011). Overvaluation and the choice of alternative earnings management mechanisms. *The Accounting Review*, *86*(5), 1491–1518, http://dx.doi.org/10.2308/accr-10092.

Balcaen, S., & Ooghe, H. (2006). 35 years of studies on business failure: an overview of the classic statistical methodologies and their related problems. *The British Accounting Review*, *38*(1), 63–93, http://dx.doi.org/10.1016/j.bar.2005.09.001.

Ball, R., & Shivakumar, L. (2005). Earnings quality in UK private firms: Comparative loss recognition timeliness. *Journal of Accounting and Economics*, *39*(1), 83–128, http://dx.doi.org/10.1016/j.jacceco.2004.04.001.

Bartov, E. (1993). The timing of asset sales and earnings manipulation. *The Accounting Review*, *68*(4), 840–855.

Beaver, W. H. (1966). Financial ratios as predictors of failure. *Journal of Accounting Research*, *4*, 71–111, http://dx.doi.org/10.2307/2490171.

Beneish, M. D. (1997). Detecting GAAP violation: implication for assessing earnings management among firms with extreme financial performance. *Journal of Accounting and Public Policy*, *16*(3), 271–309, http://dx.doi.org/10.1016/S0278-4254(97)00023-9.

Bernard, V., & Skinner, D. J. (1996). What motivates managers' choice of discretionary accruals? *Journal of Accounting and Economics*, *22*(1-3), 313–325, http://dx.doi.org/10.1016/S0165-4101(96)00431-4.

Bhojraj, S., Hribar, P., Picconi, M., & McInnis, J. (2009). Making sense of cents: An examination of firms that marginally miss or beat analyst forecasts. *Journal of Finance*, *64*(5), 2361–2388, http://dx.doi.org/10.1111/j.1540-6261.2009.01503.x.

Bhuiyan, M. H. U., & Henry, D. (2013). Real earnings management (REM) and accrual earnings management (AEM) around seasoned equity offerings (SEOs) in Australia and subsequent operating earnings performance. *Working paper*, http://dx.doi.org/10.2139/ssrn.2201973 (Accessed 5 June 2015).

Bjurman, A., & Weihagen, E. (2013). *How reliable are earnings? A study about real activities manipulation and accrual-based management in Europe.* Umea School of Business (http://www.diva-portal.org/smash/get/diva2:630890/FULLTEXT01.pdf Accessed 5 June 2015).

Blazy, R. (1999). Modifications de comportement précédant la défaillance d'entreprise: l'impact des cessions d'actifs sur la relation de crédit. *Recherches Économiques de Louvain/Louvain Economic Review*, *225–275*.

Burgstahler, D., & Dichev, I. (1997). Earnings management to avoid earnings decreases and losses. *Journal of Accounting and Economics*, *24*(1), 99–126, http://dx.doi.org/10.1016/S0165-4101(97)00017-7.

Burnett, B. M., Cripe, B. M., Martin, G. W., & McAllister, B. P. (2012). Audit quality and the trade-off between accretive stock repurchases and accrual-based earnings management. *The Accounting Review*, *87*(6), 1861–1884, http://dx.doi.org/10.2308/accr-50230.

Campa, D., & Camacho-Miñano, M. M. (2014). Earnings management among bankrupt non-listed companies: evidence from Spain. *The Spanish Review of Accounting and Finance*, *43*(1), 3–20, http://dx.doi.org/10.1080/02102412.2014.890820.

Carcello, J. V., & Neal, T. L. (2000). Audit committee composition and auditor reporting. *The Accounting Review*, *75*(4), 453–467, http://dx.doi.org/10.2308/accr.2000.75.4.453.

Carey, P., & Simnett, R. (2006). Audit partner tenure and audit quality. *The Accounting Review*, *81*(3), 653–676, http://dx.doi.org/10.2308/accr.2006.81.3.653.

Chamberlain, T., Butt, U. R., & Sarkar, S. (2014). Accrual and real earnings management around debt covenant violation. *International Advances in Economic Research*, *20*(1), 119–120, http://dx.doi.org/10.1007/s11294-013-9422-3.

Charitou, A., Lambertides, N., & Trigeorgis, L. (2007a). Earnings behaviour of financially distressed firms: the role of institutional ownership. *Abacus*, *43*(3), 271–296, http://dx.doi.org/10.1111/j.1467-6281.2007.00230.x.

Charitou, A., Lambertides, N., & Trigeorgis, L. (2007b). Managerial discretion in distressed firms. *The British Accounting Review*, *39*(4), 323–346, http://dx.doi.org/10.1016/j.bar.2007.08.003.

Chen, C. L., Yen, G., & Chang, F. H. (2009). Strategic auditor switch and financial distress prediction–empirical findings from the TSE-listed firms. *Applied Financial Economics*, *19*(1), 59–72, http://dx.doi.org/10.1080/09603100701222259.

Cheng, S. (2004). R&D expenditures and CEO compensation. *The Accounting Review*, *79*(2), 305–328, http://dx.doi.org/10.2308/accr.2004.79.2.305.

Choi, J. H., Kim, J. B., & Lee, J. J. (2011). Value relevance of discretionary accruals in the Asian financial crisis of 1997–1998. *Journal of Accounting and Public Policy*, *30*(2), 166–187, http://dx.doi.org/10.1016/j.jaccpubpol.2010.09.002.

Cohen, D. A., Dey, A., & Lys, T. Z. (2008). Real and accrual-based earnings management in the pre and post-Sarbanes-Oxley periods. *The Accounting Review*, *83*(3), 757–787, http://dx.doi.org/10.2308/accr.2008.83.3.757.

Cohen, D., Mashruwala, R., & Zach, T. (2010). The use of advertising activities to meet earnings benchmarks: Evidence from monthly data. *Review of Accounting Studies*, *15*(4), 808–832, http://dx.doi.org/10.1007/s11142-009-9105-8.

Cohen, D. A., & Zarowin, P. (2010). Accrual-based and real earnings management activities around seasoned equity offerings. *Journal of Accounting and Economics*, *50*(1), 2–19, http://dx.doi.org/10.1016/j.jacceco.2010.01.002.

Davis, E. B., & Ashton, R. H. (2002). Threshold adjustment in judgment and choice: The case of auditors' "substantial doubt" thresholds. *Organizational Behavior and Human Decision Process*, *89*(2), 1082–1099.

Dechow, P. M., Ge, W., Larson, C. R., & Sloan, R. G. (2011). Predicting material accounting misstatements. *Contemporary Accounting Research*, *28*(1), 17–82, http://dx.doi.org/10.1111/j.1911-3846.2010.01041.x.

Dechow, P. M., & Sloan, R. G. (1991). Executive incentives and the horizon problem: An empirical investigation. *Journal of Accounting and Economics*, *14*(1), 51–89, http://dx.doi.org/10.1016/0167-7187(91)90058-S.

Dechow, P., Sloan, R., & Sweeney, A. (1995). Detecting earnings management. *The Accounting Review*, *70*(2), 193–225.

DeFond, M. L., & Park, C. (2001). The reversal of abnormal accruals and the market valuation of earnings surprises. *The Accounting Review*, *76*(3), 375–404, http://dx.doi.org/10.2308/accr.2001.76.3.375.

Demers, E., & Wang, C. (2010). The impact of CEO career concerns on accruals based and real earnings management. *Working paper InSEAD*, http://dx.doi.org/10.2139/ssrn.1562428 (Accessed 16 June 2015).

Dewaelheyns, N., & Van Hulle, C. (2008). Legal reform and aggregate small and micro business bankruptcy rates: evidence from the 1997 Belgian bankruptcy code. *Small Business Economics*, *31*(4), 409–424, http://dx.doi.org/10.1007/s11187-007-9060-3.

Durnev, A. (2010). The real effects of political uncertainty: Elections and investment sensitivity to stock prices. *Working paper*, http://dx.doi.org/10.2139/ssrn.1695382 (Accessed 24 June 2015).

Eisele, A. (2012). *Target Shooting? Benchmark-driven earnings management in Germany*. (Doctoral dissertation) (http://verdi.unisg.ch/www/edis.nsf/SysLkpByIdentifier/4009/$FILE/dis4009.pdf Accessed 21 June 2015).

Enomoto, M., Kimura, F., & Yamaguchi, T. (2015). Accrual-based and real earnings management: an international comparison for investor protection. *Journal of Contemporary Accounting and Economics*, http://dx.doi.org/10.2139/ssrn.2066797 (in press, Accessed 24 June 2015).

European Commission (2014). A partial and fragile recovery. Annual Report on SMEs 2013/2014. http://ec.europa.eu/enterprise/policies/sme/facts-figures-analysis/performance-review/files/supporting-documents/2014/annual-report-smes-2014_en.pdf (Accessed 24 June 2015)

Ewert, R., & Wagenhofer, A. (2005). Economic effects of tightening accounting standards to restrict earnings management. *Accounting Review*, *80*(4), 1101–1124, http://dx.doi.org/10.2308/accr.2005.80.4.1101.

Fich, E. M., & Slezak, S. L. (2008). Can corporate governance save distressed firms from bankruptcy? An empirical analysis. *Review of Quantitative Finance and Accounting*, *30*(2), 225–251, http://dx.doi.org/10.1007/s11156-007-0048-5.

García Lara, J. M., García Osma, B., & Mora, A. (2005). The effect of earnings management on the asymmetric timeliness of earnings. *Journal of Business Finance and Accounting*, *32*(3-4), 691–726.

García Lara, J. M., García Osma, B., & Neophytou, E. (2009). Earnings quality in ex-post failed firms. *Accounting and Business Research*, *39*(2), 119–138, http://dx.doi.org/10.1080/00014788.2009.9663353.

Geertsema, P. G., Lont, D. H., & Lu, H. (2014). Earnings management around CEO turnovers. *Working paper*, http://dx.doi.org/10.2139/ssrn.2517013 (Accessed 4 June 2015).

Geiger, M., & van der Laan Smith, J. (2010). The effect of institutional and cultural factors on the perceptions of earnings management. *Journal of International Accounting Research*, *9*(2), 21–43, http://dx.doi.org/10.2308/jiar.2010.9.2.21.

Graham, J. R., Harvey, C. R., & Rajgopal, S. (2005). The economic implications of corporate financial reporting. *Journal of Accounting and Economics*, *40*(1-3), 3–73.

Grice, J. S., & Ingram, R. W. (2001). Tests of the generalizability of Altman's bankruptcy prediction model. *Journal of Business Research*, *54*(1), 53–61, http://dx.doi.org/10.1016/S0148-2963(00)00126-0.

Guay, W. R., Kothari, S. P., & Watts, R. L. (1996). A market-based evaluation of discretionary accrual models. *Journal of Accounting Research*, *34*, 83–105, http://dx.doi.org/10.2307/2491427.

Gunny, K. A. (2010). The relation between earnings management using real activities manipulation and future performance: evidence from meeting earnings benchmarks. *Contemporary Accounting Research*, *27*(3), 855–888, http://dx.doi.org/10.1111/j.1911-3846.2010.01029.x.

Habib, A., Uddin Bhuiyan, B., & Islam, A. (2013). Financial distress, earnings management and market pricing of accruals during the global financial crisis. *Managerial Finance*, *39*(2), 155–180, http://dx.doi.org/10.1108/03074351311294007.

Healy, P. (1996). Discussion of a market-based evaluation of discretionary accrual models. *Journal of Accounting Research*, *34*, 107–115, http://dx.doi.org/10.2307/2491428.

Herrmann, D., Inoue, T., & Thomas, W. B. (2003). The sale of assets to manage earnings in Japan. *Journal of Accounting Research*, *41*(1), 89–108, http://dx.doi.org/10.1111/1475-679X.00097.

Hewitt, M., Hodge, F., & Pratt, J. (2014). How the discovery of accruals-based vs. real earnings management affects investment decisions. *Working paper*, http://dx.doi.org/10.2139/ssrn.2245204 (Accessed 4 June 2015).

Iatridis, G., & Kadorinis, G. (2009). Earnings management and firm financial motives: A financial investigation of UK listed firms. *International Review of Financial Analysis*, *18*(4), 164–173, http://dx.doi.org/10.1016/j.irfa.2009.06.001.

Ipino, E., & Parbonetti, A. (2011). Mandatory IFRS adoption: the trade-off between accrual and real-based earnings management. *Working paper*, http://dx.doi.org/10.2139/ssrn.2039711 (Accessed 4 June 2015).

Jackson, S. B., & Wilcox, W. E. (2000). Do managers grant sales price reductions to avoid losses and declines in earnings and sales? *Quarterly Journal of Business and Economics*, *39*(4), 3–20.

Jaggi, B., & Lee, P. (2002). Earnings management response to debt covenant violations and debt restructuring. *Journal of Accounting, Auditing and Finance*, *17*(4), 295–324, http://dx.doi.org/10.1177/0148558X0201700402.

Jiang, F., Zhu, B., & Huang, J. (2013). CEO's financial experience and earnings management. *Journal of Multinational Financial Management*, *23*(3), 134–145, http://dx.doi.org/10.1016/j.mulfin.2013.03.005.

Jones, J. (1991). Earnings management during import relief investigations. *Journal of Accounting Research*, *29*(2), 193–228, http://dx.doi.org/10.2307/2491047.

Kasznik, R. (1999). On the association between voluntary disclosure and earnings management. *Journal of Accounting Research*, *37*(1), 57–81, http://dx.doi.org/10.2307/2491396.

Kennedy, P. (2008). *A Guide to Econometrics*. Hoboken, NJ: Wiley-Blackwell.

Kim, J. B., Chung, R., & Firth, M. (2003). Auditor conservatism, asymmetric monitoring, and earnings management. *Contemporary Accounting Research*, *20*(2), 323–359, http://dx.doi.org/10.1506/J29K-MRUA-0APP-YJ6V.

Kim, Y. S., Kim, Y., & Song, K. (2013). Credit rating changes and earnings management. *Asia-Pacific Journal of Financial Studies*, *42*(1), 109–140, http://dx.doi.org/10.1111/ajfs.12007.

Kim, J. B., & Sohn, B. C. (2013). Real earnings management and cost of capital. *Journal of Accounting and Public Policy*, *32*(6), 518–543, http://dx.doi.org/10.1016/j.jaccpubpol.2013.08.002.

Kothari, S. P., Leone, A., & Wasley, C. (2005). Performance matched discretionary accrual measures. *Journal of Accounting and Economics*, *39*(1), 163–197, http://dx.doi.org/10.1016/j.jacceco.2004.11.002.

Kothari, S. P., Roychowdhury, S., & Mizik, N. (2015). Managing for the moment: the role of real activity versus accruals earnings management in SEO valuation. *The Accounting Review*, http://dx.doi.org/10.2139/ssrn.1982826 (in press).

Krishnan, J., & Krishnan, J. (1997). Litigation risk and auditor resignations. *Accounting Review*, *72*(4), 539–560.

Laitinen, E. (2011). Extension of break-even analysis for payment default prediction: evidence from small firms. *Investment Management and Financial Innovations*, *8*(4), 96–108.

Lenard, M. J., Petruska, K. A., Alam, P., & Yu, B. (2013). Internal control weaknesses and evidence of real activities manipulation. *Working paper*, http://dx.doi.org/10.2139/ssrn.2367334 (Accessed 4 June 2015).

Leuz, C., Nanda, D., & Wysocki, P. (2003). Earnings management and investor protection: an international comparison. *Journal of Financial Economics*, *69*(3), 505–527, http://dx.doi.org/10.1016/S0304-405X(03)00121-1.

Liu, X., Hodgkinson, I. R., & Chuang, F. (2014). Foreign competition, domestic knowledge base and innovation activities: Evidence from Chinese high-tech industries. *Research Policy*, *43*(2), 414–422, http://dx.doi.org/10.1016/j.respol.2013.11.005.

Matsuura, S. (2008). On the relation between real earnings management and accounting earnings management: Income smoothing perspective. *Journal of International Business Research*, *7*(3), 63–82.

Mousavi, M. M., Ouenniche, J., & Xu, B. (2015). Performance evaluation of bankruptcy prediction models: An orientation-free super-efficiency DEA-based framework. *International Review of Financial Analysis*, *42*, 64–75.

Nabar, S., & Boonlert-U-Thai, K. K. (2007). Earnings management, investor protection and national culture. *Journal of International Accounting Research*, *6*(2), 35–54, http://dx.doi.org/10.2308/jiar.2007.6.2.35.

Nelson, M. W., Elliot, J. A., & Tarpley, R. L. (2002). Evidence from auditors about managers' and auditors' earnings-management decisions. *The Accounting Review*, *77*(1), 157–202, http://dx.doi.org/10.2308/accr.2002.77.s-1.175.

Park, Y., & Shin, H. (2004). Board composition and earnings management in Canada. *Journal of Corporate Finance*, *10*(3), 431–457, http://dx.doi.org/10.1016/S0929-1199(03)00025-7.

Perry, S., & Grinaker, R. (1994). Earnings expectations and discretionary research and develop. *Accounting Horizons*, *8*(4), 43.

Pindado, J., Rodrigues, L., & de la Torre, C. (2006). How does financial distress affect small firms' financial structure? *Small Business Economics*, *26*(4), 377–391, http://dx.doi.org/10.1007/s11187-005-4845-8.

Ricci, C. W. (2011). Manipulating receivables: a comparison using the SEC's accounting auditing enforcement releases. *Journal of Applied Business and Economics*, *12*(5), 35–44.

Rosner, R. L. (2003). Earnings manipulation in failing firms. *Contemporary Accounting Research*, *20*(2), 361–408, http://dx.doi.org/10.1506/8EVN-9KRB-3AE4-EE81.

Roychowdhury, S. (2006). Earnings management through real activities manipulation. *Journal of Accounting and Economics*, *42*(3), 335–370, http://dx.doi.org/10.1016/j.jacceco.2006.01.002.

Schipper, K. (1989). Commentary on earnings management. *Accounting Horizons*, *3*(4), 91–102.

Shan, Y., Taylor, S., & Walter, T. (2013). Error in estimating unexpected accruals in the presence of large changes in net external financing. *Working paper*, http://dx.doi.org/10.2139/ssrn.1572164 (Accessed 14 June 2015).

Sweeney, A. P. (1994). Debt-covenant violations and managers' accounting responses. *Journal of Accounting and Economics*, *17*(3), 281–308.

Teoh, S. H., Welch, I., & Wong, T. J. (1998). Earnings management and the underperformance of seasoned equity offerings. *Journal of Financial Economics*, *50*(1), 63–99, http://dx.doi.org/10.1016/S0304-405X(98)00032-4.

Tinoco, M. H., & Wilson, N. (2013). Financial distress and bankruptcy prediction among listed companies using accounting, market and macroeconomic variables.

International Review of Financial Analysis, *30*, 394–419, http://dx.doi.org/10.1016/j.irfa.2013.02.013.

Trombetta, M., & Imperatore, C. (2014). The dynamic of financial crises and its non-monotonic effects on earnings quality. *Journal of Accounting and Public Policy*, *33*(3), 205–232, http://dx.doi.org/10.1016/j.jaccpubpol.2014.02.002.

Tykvová, T., & Borell, M. (2012). Do private equity owners increase risk of financial distress and bankruptcy? *Journal of Corporate Finance*, *18*(1), 138–150, http://dx.doi.org/10.1016/j.jcorpfin.2011.11.004.

Wongsunwai, W. (2012). The effect of external monitoring on accrual-based and real earnings management: evidence from venture-backed initial public offerings. *Contemporary Accounting Research*, *30*(1), 296–324, http://dx.doi.org/10.1111/j.1911-3846.2011.01155.x.

Wooldridge, J. (2002). *Econometric analysis of cross section and panel data.* Cambridge, Massachusetts: The MITPress.

Xu, R. Z., Taylor, G. K., & Dugan, M. T. (2007). Review of real earnings management literature. *Journal of Accounting Literature*, *26*, 195–228.

Yang, L., Rahman, A., & Bradbury, M. (2015). The trade-off between real earnings management and accruals management. *Working paper* (http://www.researchgate.net/publication/228976807. Accessed 24 June 2015).

Zang, A. Y. (2012). Evidence on the trade-off between real activities manipulation and accrual-based earnings management. *The Accounting Review*, *87*(2), 675–703, http://dx.doi.org/10.2308/accr-10196.

Zmijewski, M. E. (1984). Methodological issues related to the estimation of financial distress prediction models. *Journal of Accounting Research*, *22*, 59–82, http://dx.doi.org/10.2307/2490859.